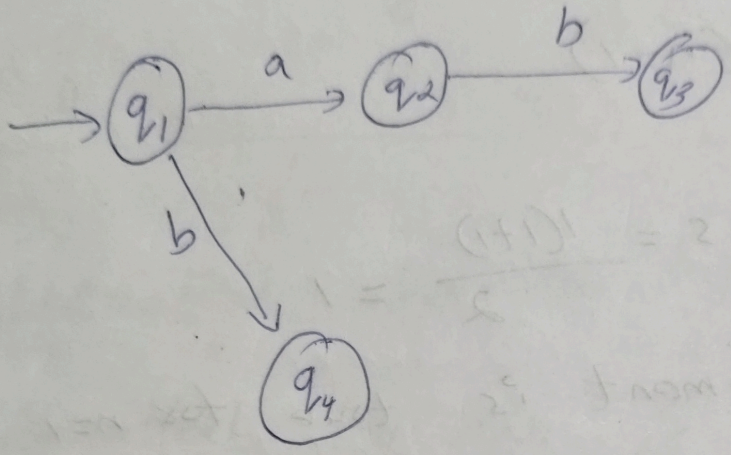
Test - CO1-AT2

Identify FA and NFA

i)



Ans:

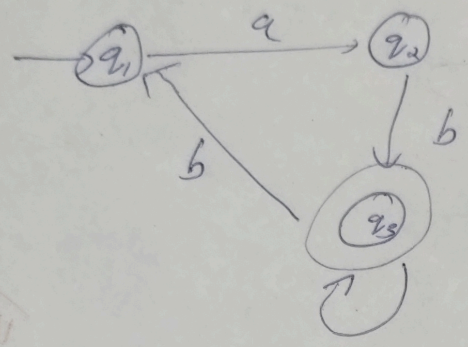
Transitions:

* Start state = q_1
* From q_1, input a goes to q_2
* From q_2, input b goes to q_3
* From q_1, input b goes to q_4

Input State	Current State	Input Symbol	Next State
q_1	q_1	a	q_2
q_1	q_1	b	q_4
q_2	q_2	b	q_3

The state q_1 has two outgoing transitions one on a and one on b, this is actually allowed in FA because the inputs are different, so, based on the diagram shown it is FA.

i)



transitions;

start state: q_1

$q_1 \xrightarrow{a} q_2$

$q_2 \xrightarrow{b} q_3$

$q_3 \xrightarrow{b} q_1$

Input	Current	Input Symbol	Next state
q_1	q_1	a	q_2
q_2	q_2	b	q_3
q_3	q_3	b	q_1

FA, each state has at most one transition for each input symbol.